National College of Ireland

Automated Gaelic Football Match Analysis using Computer Vision and Deep Learning

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1.0 Objectives

This project sets out to achieve the development of an intelligent, automated system for analysing Gaelic Football matches from 2D video footage. The primary objective is to leverage computer vision and deep learning techniques to extract meaningful tactical and performance-based data, which is currently a largely manual and time-consuming process for most clubs.

The core aims of the project are:

1. **Kick Detection:** To develop and train a deep learning model capable of accurately identifying and timestamping the action of a player kicking the football within a video frame.
2. **Ball Tracking and Trajectory Analysis:** To implement a robust tracking algorithm that follows the football's trajectory immediately after a kick, calculating its path and speed.
3. **Shot Outcome Classification:** To create a classification module that automatically determines the outcome of a shot on goal, categorising it as a point, a wide, or short/blocked.
4. **Structured Data Generation:** To process the raw video analysis into a structured format (CSV/JSON). This output will contain quantitative metrics such as the timestamp of each kick, shot outcome, and trajectory coordinates, forming a foundation for advanced analytics.
5. **System Scalability:** To design the system architecture with future scalability in mind, enabling potential expansion to include live video analysis, player identification, and integration with a performance dashboard on a cloud platform like Azure or AWS.

Ultimately, this project aims to deliver a proof-of-concept tool that demonstrates the feasibility of applying modern AI techniques to Gaelic Football, providing valuable, data-driven insights for coaches and players.

2.0 Background

Gaelic Football is one of Ireland's most popular and culturally significant sports, yet it lags behind other major sports like soccer and basketball in the adoption of advanced data analytics. Performance analysis at amateur and semi-professional levels often relies on manual video review, which is subjective, labour-intensive, and prone to human error. This creates a significant opportunity to introduce technology that can provide objective, detailed, and efficient analysis.

My motivation for this project stems from a passion for both sports and data science. There is a clear gap in the market for an accessible, AI-powered analytics tool tailored specifically for the nuances of Gaelic Football. This project will meet the objectives outlined in Section 1.0 by systematically building a modular system. I will start by researching and mastering relevant technologies like YOLOv8 and OpenCV. Following this, I will develop each core

module—kick detection, ball tracking, and outcome classification—sequentially. By focusing on these fundamental events, the project will create a robust foundation that automates the most critical aspects of match analysis, paving the way for more complex features in the future and providing tangible value to the Gaelic Games community.

3.0 State of the Art

The field of sports analytics has seen significant advancements through computer vision. Companies like **Hudl** and **Stats Perform** offer sophisticated video analysis platforms for professional sports like soccer, basketball, and American football. Academically, numerous studies have applied deep learning for tasks such as player tracking, action recognition, and ball tracking. For instance, research has successfully used Convolutional Neural Networks (CNNs) and models like YOLO for real-time object detection in soccer to identify players and the ball. Similarly, tracking algorithms such as SORT (Simple Online and Realtime Tracking) have been adapted for tracking multiple players simultaneously.

However, the vast majority of this research and commercial development is focused on globally popular sports. **Gaelic Football remains a largely unexplored domain in computational analysis.** This project stands out by being one of the first to apply state-of-the-art deep learning models specifically to this sport.

My work differs from existing solutions in three key ways:

1. **Sport-Specific Application:** It is tailored to the unique actions of Gaelic Football, such as the solo and the distinct kicking style, which are not present in other sports.
2. **Creation of a Novel Dataset:** A core contribution of this project will be the creation of a custom, annotated video dataset of Gaelic Football actions. This dataset is essential for training accurate models and does not currently exist.
3. **Accessibility:** While professional tools are expensive and complex, this project aims to build a foundational system that could eventually be developed into a more accessible tool for local clubs and academies with limited resources.

4.0 Data

The primary data required for this project is 2D video footage of Gaelic Football matches. The system will be designed to work with standard broadcast or recording quality, making it versatile for various sources.

Data Sourcing:

The initial dataset will be compiled from publicly available sources. I will download full-length matches and highlight reels from platforms like YouTube, RTE Player, and GAA GO. The selection will aim for a variety of camera angles (primarily wide, broadcast-style shots), lighting conditions (daylight, floodlit), and match types (club, inter-county) to ensure the model's robustness and generalisability. I anticipate collecting approximately 5-10 hours of raw video footage to create the initial dataset.

Data Preparation and Annotation:

As no pre-existing annotated dataset for Gaelic Football actions exists, a significant part of this project will involve manual data annotation. This is a critical step for training a supervised deep learning model.

The annotation process will involve:

1. **Frame Extraction:** Using a script (e.g., Python with OpenCV), I will extract frames from the collected video clips that contain the actions of interest.
2. **Annotation:** I will use an annotation tool such as **CVAT (Computer Vision Annotation Tool)** or **Roboflow** to manually label objects in these frames. The annotations will consist of:
    - **Bounding Boxes:** A box will be drawn around the player in the act of kicking the ball. This will be labelled kick.
    - **Ball Detection:** A separate bounding box will be drawn around the football, labelled ball.
    - **Outcome Labelling:** The video clips containing shots will be categorised and labelled as point, wide, or short. This label will be used for training the outcome classification logic.

I aim to generate an initial dataset of approximately **1,000-2,000 annotated frames**. This dataset will be split into training, validation, and testing sets (e.g., 70%, 15%, 15% split) to train the model and evaluate its performance effectively. This custom-built, annotated dataset will be one of the key deliverables of this project.

5.0 Methodology & Analysis

The project will follow a structured, modular development methodology, allowing for iterative development, testing, and refinement of each component before integration. The overall approach can be broken down into four main phases.

Phase 1: Data Collection and Pre-processing (Weeks 1-4)

- **Task:** Gather Gaelic Football match footage from online sources.
- **Task:** Extract relevant clips and frames containing kicking actions and shots at goal.
- **Task:** Manually annotate the extracted frames with bounding boxes for 'kick' and 'ball' using CVAT.
- **Milestone:** A well-annotated and structured dataset, split into training, validation, and test sets.

Phase 2: Model Development and Training (Weeks 5-12)

- **Task:** Develop and train the Kick Detection model using the YOLOv8 architecture on the custom dataset.

- **Task:** Develop and train a separate Ball Detection model, also using YOLOv8, fine-tuned for detecting a small, fast-moving object.
- **Task:** Implement a ball tracking algorithm (e.g., Kalman Filter or OpenCV's tracker library) that takes the output of the ball detection model to predict its trajectory across frames.
- **Milestone:** Trained models for kick detection and ball detection, and a functional ball tracking module.

Phase 3: System Integration and Classification (Weeks 13-18)

- **Task:** Integrate the individual modules into a single processing pipeline. The pipeline will trigger ball tracking upon the detection of a 'kick' event.
- **Task:** Develop the logic for Shot Outcome Classification. This will be a rule-based system that analyses the final coordinates of the ball's trajectory relative to the goalposts' location (which may need to be manually defined or detected per video).
- **Task:** Implement the data output functionality, where the system logs event details (timestamp, event type, trajectory data, outcome) into a structured CSV or JSON file.
- **Milestone:** A complete, end-to-end prototype system capable of processing a video file and generating an analysis report.

Phase 4: Evaluation and Documentation (Weeks 19-24)

- **Task:** Evaluate the system's performance using the test dataset. Metrics will include precision, recall, and F1-score for kick detection, and overall accuracy for outcome classification.
- **Task:** Refine the models and algorithms based on the evaluation results.
- **Task:** Write the final project report, detailing the system architecture, methodology, results, and potential future work.
- **Milestone:** A final, evaluated prototype and comprehensive project documentation.

This phased approach ensures a logical progression from data to a fully functional system, with clear milestones to track progress.

6.0 Technical Details

This project will be developed primarily in **Python**, leveraging its extensive ecosystem of libraries for computer vision and machine learning. The initial development and testing will be performed on a local machine with a suitable GPU to accelerate model training. The architecture is designed to be cloud-agnostic for future deployment on platforms like AWS or Azure.

Core Technologies and Algorithms:

- **Programming Language:** Python 3.8+

- **Core Libraries:**
    - **OpenCV:** For all fundamental video processing tasks, including reading/writing video files, frame extraction, and image manipulation. It will also be used for implementing tracking algorithms.
    - **PyTorch:** The deep learning framework on which YOLOv8 is built. It will be used for training and fine-tuning the detection models.
    - **NumPy & Pandas:** For numerical operations, data manipulation, and handling the structured CSV/JSON outputs.

Key Algorithms and Models:

1. **Object Detection (Kick & Ball Detection):**
    - **YOLOv8 (You Only Look Once v8):** This state-of-the-art, real-time object detection model will be the core of the detection modules. It is chosen for its high accuracy and impressive inference speed, which is crucial for video analysis. I will use a pre-trained YOLOv8 model and fine-tune it on my custom-annotated Gaelic Football dataset. This transfer learning approach will significantly reduce training time and improve performance.
2. **Ball Tracking:**
    - **Kalman Filter or SORT (Simple Online and Realtime Tracking):** Once the ball is detected in a frame, a tracking algorithm is needed to follow it in subsequent frames, especially if detection fails intermittently. A Kalman Filter is a strong candidate as it can predict the ball's position based on its past motion, making the tracking robust. SORT is another powerful alternative that associates detections across frames.
3. **Outcome Classification Logic:**
    - This will likely be a **rule-based algorithm**. After a shot is detected, the ball's trajectory will be tracked. The final position of the trajectory will be compared against the coordinates of the goalposts.
    - The goalpost locations can be initially defined manually for each video clip.
    - The logic will check:
        - Did the ball's trajectory pass between the vertical posts and over the crossbar? -> **Point**
        - Did the ball's trajectory pass outside the vertical posts? -> **Wide**
        - Did the ball's trajectory end significantly before reaching the goal line? -> **Short**

The overall workflow will be orchestrated by a main Python script that takes a video file path as input, processes it frame-by-frame through the detection and tracking pipeline, and outputs the final analysis file.

7.0 Project Plan

Phase	Task	Weeks	Deliverable
Phase 1: Research & Planning	**1.1:** Literature Review on Sports CV	1-2	Summary of relevant papers and techniques.
	1.2: Finalise Tech Stack (YOLOv8, OpenCV)	2	Confirmed list of tools and libraries.
	1.3: Detailed Project Scoping and Timeline	3	Refined Gantt chart and project goals.
Phase 2: Data Acquisition & Prep	**2.1:** Collect Video Footage (YouTube, etc.)	3-4	5-10 hours of raw match footage.
	2.2: Set up Annotation Environment (CVAT)	4	Functional annotation workspace.
	2.3: Frame Extraction and Annotation	5-8	~1,500+ annotated frames for training.
	2.4: Dataset Splitting and Formatting	8	Formatted training, validation, & test sets.
Phase 3: Module Development	**3.1:** Kick Detection Model Training	9-12	Fine-tuned YOLOv8 model for kick detection.
	3.2: Ball Detection Model Training	11-14	Fine-tuned YOLOv8 model for ball detection.

	3.3: Implement Ball Tracking Algorithm	13-16	Functional ball trajectory tracking script.
	3.4: Develop Outcome Classification Logic	15-18	Script to classify shots based on trajectory.
Phase 4: Integration & Testing	**4.1:** Build End-to-End Processing Pipeline	17-19	Integrated system script.
	4.2: System Testing with unseen video data	19-20	Initial performance metrics (accuracy, etc.).
	4.3: Performance Evaluation & Model Tuning	20-22	Evaluation report and improved models.
	4.4: Implement Data Output (CSV/JSON)	21-22	Function to save analysis results.
Phase 5: Documentation & Finalisation	**5.1:** Write Mid-Point Project Report	12	Mid-point documentation submission.
	5.2: Write Final Project Report & Thesis	22-24	Complete project dissertation.
	5.3: Prepare Final Presentation & Demo	23-24	Project presentation slides and video demo.
	5.4: Code Cleanup and Repository Finalisation	24	Well-documented code on GitHub.

Gantt Chart Representation:

Task	W1-4	W5-8	W9-12	W13-16	W17-20	W21-24
1. Research & Plan	■■■■					
2. Data Prep		■■■■■				
3. Model Dev			■■■■	■■		
4. Tracking Dev				■■■■		
5. Integration					■■■■	
6. Evaluation					■■	■■
7. Documentation			■■			■■■■